GCSE PHYSICS · TEACHING DECK

Energy Stores & Changes

Deck 2 of 4 — Kinetic & gravitational potential energy

AQA 8463 | Edexcel 1PH0 | OCR J249 / J250

The two most-tested calculations in the whole topic — drill selecting and rearranging the equation



Where we're going — 4 decks



1

Systems, transfers & how to describe them

All tiers

YOU ARE HERE

2

Kinetic & gravitational potential energy

All tiers

3

Elastic potential energy & bringing it together

Higher stretch

4

Quick recap & multi-step challenges

Review

Still all tiers, all boards — Foundation and Higher take these two equations at the same pace, just go steady on the rearranging.

STARTER *~1 min*



"Two cars at the same speed, one twice the mass — which has more kinetic energy, and how many times more? Now same mass but one at twice the speed — how many times more?"

Push toward: Mass doubles → KE doubles. Speed doubles → KE is four times bigger, because speed is squared.

Tutor: *Draws out the key idea before the formal equation: speed is squared, mass isn't.*

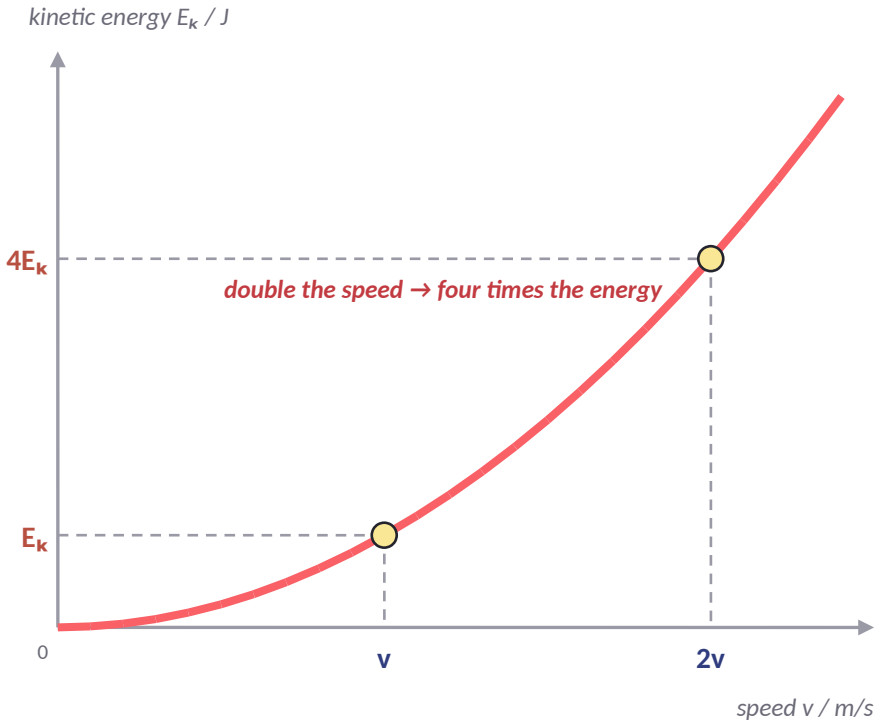
THEIR IDEAS

Kinetic energy — $E_k = ½ m v^2$

- Kinetic energy is the energy in the store of a moving object.
- $E_k = ½ m v^2$, where E_k = kinetic energy (J), m = mass (kg), v = speed (m/s).
- Speed is squared, so doubling the speed gives four times the kinetic energy — this is why stopping distances grow so quickly.
- Rearranged for speed: $v = \sqrt{(2E_k \div m)}$.

$E_k = ½ m v^2$

J = 0.5 × kg × (m/s)² · rearranged for speed: $v = \sqrt{(2E_k \div m)}$



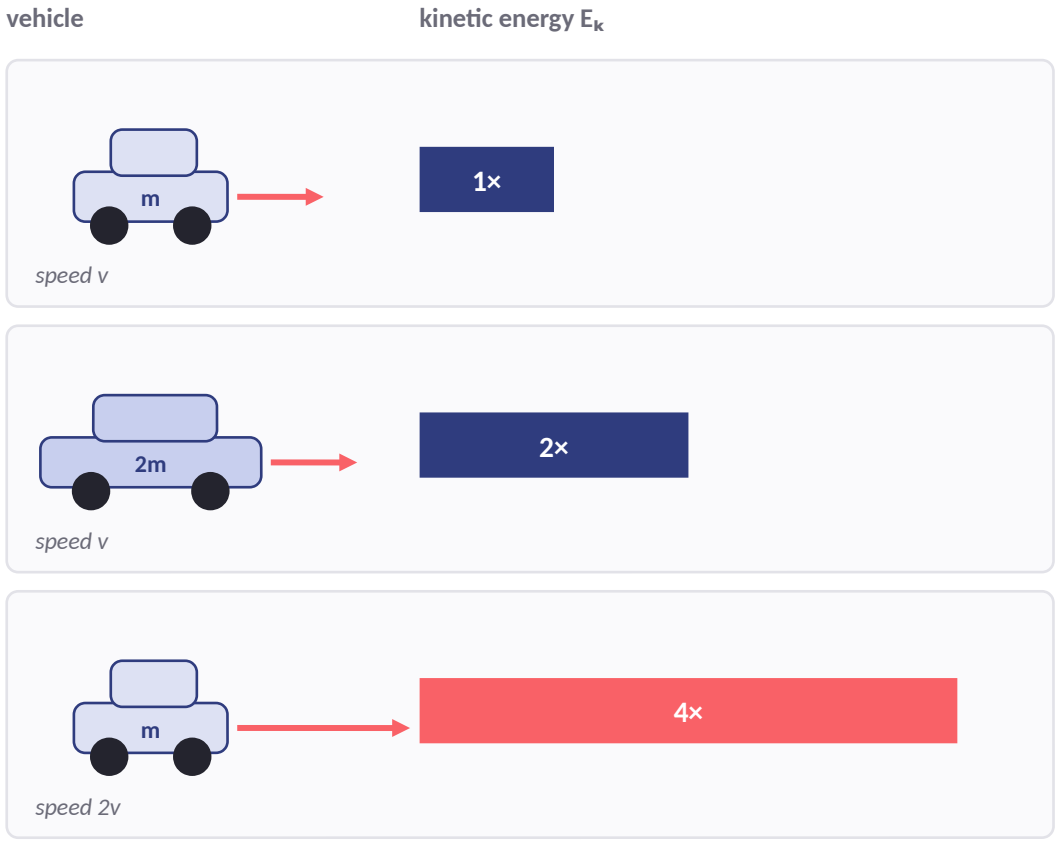
Doubling v quadruples E_k — it's squared in the equation.

A formula triangle can't rearrange a squared term — practise rearranging directly.

Where marks are lost: forgetting to square the speed (square only v, not ½mv); dropping the ½; forgetting to square-root at the end when finding v.

Mass or speed: which matters more?

- Double the mass → double the kinetic energy (E_k is directly proportional to m).
- Double the speed → four times the kinetic energy (E_k is proportional to v^2). Treble the speed → nine times.
- **Rearranging: $E_k = ½ m v^2$ → $v^2 = 2E_k ÷ m$ → $v = √(2E_k ÷ m)$. Mass: $m = 2E_k ÷ v^2$.**
- Convert first: speed in m/s (km/h ÷ 3.6), mass in kg (g ÷ 1000, tonnes × 1000).



Double the mass → ×2. Double the speed → ×4.

$$v = √(2E_k ÷ m)$$

m/s = √(J ÷ kg) · square-root as the very last step

Where marks are lost: converting km/h and g after substituting; squaring the whole of ½mv; reading "×4" as "×2".

WORKED EXAMPLE

Tutor models the method — answer comes last

A 1200 kg car travels at 20 m/s. Find its kinetic energy.

METHOD

- Write the equation: $E_k = ½ m v^2$
- Substitute: $E_k = ½ \times 1200 \times 20^2$
- Square the speed first: $20^2 = 400$, then multiply through.

WORK IT THROUGH

Answer: $E_k = ½ \times 1200 \times 400 = 240\,000$ J (240 kJ).

Mark scheme: *1 for equation · 1 for substitution (squared speed) · 1 for 240 000 J.*

WORKED EXAMPLE — FINDING SPEED

Rearrange first — then square-root last

A 0.40 kg football has 20 J of kinetic energy. Find its speed.

METHOD

- Write the equation: $E_k = \frac{1}{2} m v^2$
- Rearrange for v^2: $v^2 = 2E_k \div m = (2 \times 20) \div 0.40$
- Calculate $v^2 = 40 \div 0.40 = 100$, then take the square root.

WORK IT THROUGH

Answer: $v = \sqrt{100} = 10$ m/s.

Mark scheme: *1 for rearranging · 1 for substituting ($v^2 = 100$) · 1 for v = 10 m/s.*

YOUR TURN — PART 1

Kinetic energy ($E_k = ½ m v^2$) — answers on the next slide.

1 State the kinetic energy equation with units.

2 A 0.15 kg ball moves at 12 m/s. Calculate its kinetic energy.

3 A 70 kg runner has 1260 J of kinetic energy. Calculate their speed.

4 A car doubles its speed. By what factor does its kinetic energy change?

Working space is squared on purpose — students calculate, tutor annotates.

ANSWERS — PART 1

Tutor reveal

1

$E_k = \frac{1}{2} m v^2$ (J, kg, m/s).

2

$\frac{1}{2} \times 0.15 \times 12^2 = 0.075 \times 144 = 10.8$ J.

3

$v = \sqrt{(2 \times 1260 \div 70)} = \sqrt{36} = 6$ m/s.

4

×4 (energy is proportional to speed^2).

STARTER *~1 min*



"Why does a dropped brick hurt far more from a second-floor window than from a chair?"

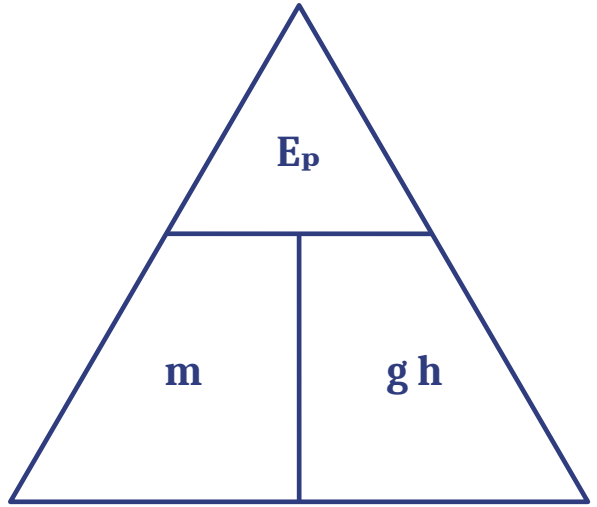
Push toward: Raising something stores more gravitational potential energy — more height, more energy to release.

Tutor: *Straightforward once the value of g is clear — this is the gentler of the two calculations in this deck.*

THEIR IDEAS

Gravitational potential energy — $E_p = m\ g\ h$

- Gravitational potential energy is the energy in the store of an object raised in a gravitational field.
- E_p = m g h, where E_p = gravitational potential energy (J), m = mass (kg), g = gravitational field strength (N/kg), h = height (m). Edexcel writes this ΔGPE = m g Δh — the same equation.
- On Earth g ≈ 9.8 N/kg. Some questions use 10 N/kg — always use the value given. Only the change in height matters.



Cover the quantity you want: $m = E_p \div (g\ h)$; $h = E_p \div (m\ g)$.

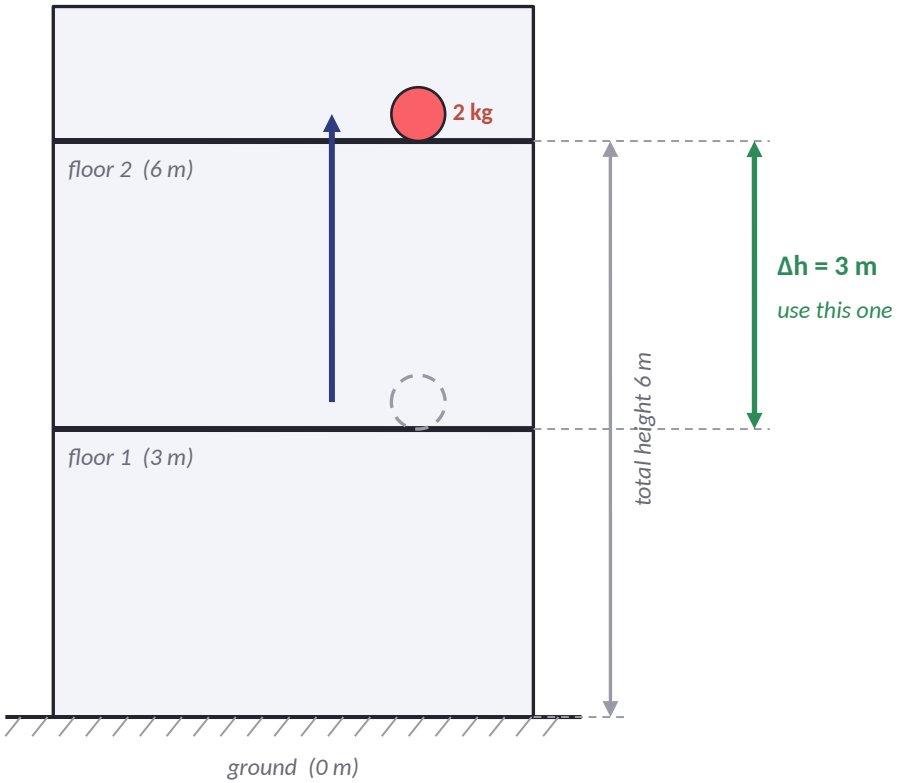
$E_p = m\ g\ h$

J = kg × N/kg × m · g ≈ 9.8 N/kg (or 10 N/kg if the question says so)

Where marks are lost: using the wrong g; using the total height instead of the change in height; mixing up mass (kg) and weight (N).

Change in height, weight and the value of g

- Only the CHANGE in height counts: a ball lifted from floor 1 (3 m) to floor 2 (6 m) gains E_p for Δh = 3 m, not 6 m.
- Weight W = m g (N). So E_p = W × h — the work done lifting the object through the height.
- **Mass is in kg, weight is in N. If the question gives a weight, divide by g to get the mass — or use E_p = W × h directly.**
- Edexcel: ΔGPE = m g Δh. AQA / OCR: E_p = m g h. Same equation — and g = 9.8 or 10 N/kg, whichever the question states.



Raising the 2 kg ball from floor 1 to floor 2: E_p = m g Δh = 2 × 10 × 3 = 60 J

Where marks are lost: using 6 m instead of 3 m; converting cm to m after multiplying; writing the answer in N instead of J.

WORKED EXAMPLE

Tutor models the method — answer comes last

Lift a 3 kg bag 1.5 m onto a shelf (g = 9.8). GPE gained?

METHOD

- Write the equation: $E_p = m\ g\ h$
- Substitute: $E_p = 3 \times 9.8 \times 1.5$

WORK IT THROUGH

Answer: $E_p = 3 \times 9.8 \times 1.5 = 44.1$ J.

Mark scheme: *1 for equation · 1 for substitution · 1 for 44.1 J.*

WORKED EXAMPLE — FINDING HEIGHT

Rearrange for h, then substitute

A 0.20 kg ball gains 3.0 J of gravitational potential energy (g = 10 N/kg). How high was it raised?

METHOD

- Write the equation: $E_p = m\ g\ h$
- Rearrange for h: $h = E_p \div (m\ g)$
- Substitute: $h = 3.0 \div (0.20 \times 10) = 3.0 \div 2.0$

WORK IT THROUGH

Answer: h = 1.5 m.

Mark scheme: *1 for rearranging · 1 for substitution · 1 for 1.5 m.*

YOUR TURN — PART 2

Gravitational potential energy ($E_p = m\,g\,h$) — answers on the next slide.

1 State the GPE equation with units.

2 A 60 kg climber ascends 20 m (g = 9.8). GPE gained?

3 A 0.5 kg ball gains 24.5 J of GPE. How high was it raised (g = 9.8)?

4 Using g = 10, find the GPE of a 2 kg mass 5 m up.

Working space is squared on purpose — students calculate, tutor annotates.

ANSWERS — PART 2

Tutor reveal

1

$E_p = m\ g\ h$ (J, kg, N/kg, m).

2

60 × 9.8 × 20 = 11 760 J.

3

h = 24.5 ÷ (0.5 × 9.8) = 5 m.

4

2 × 10 × 5 = 100 J.

EXAM PRACTICE — SET A

Kinetic energy · 15 marks in total · marks shown in [brackets] · mark scheme on the next slide

Q1 *Calculation · multi-step* **[8 marks]**

A car of mass 800 kg is travelling at 90 km/h along a motorway.

(a) Show that 90 km/h is the same as 25 m/s. [1]

(b) Calculate the kinetic energy of the car. [3]

(c) The car slows to 45 km/h. Explain why its kinetic energy is not half of the value in (b). [2]

(d) The car brakes to rest. State the store the energy is transferred to and the pathway. [2]

Q2 *Data table · calculation* **[7 marks]**

A student measures the speed of a 0.50 kg trolley and calculates its kinetic energy.

speed / m/s	1.0	2.0	3.0	4.0
kinetic energy / J	0.25	1.0		4.0

(a) Calculate the kinetic energy of the trolley at 3.0 m/s. [2]

(b) The student says: “Kinetic energy is directly proportional to speed.” Use the table to show that this is wrong. [2]

(c) Calculate the speed of the trolley when its kinetic energy is 8.0 J. Give your answer to 2 significant figures. [3]

MARK SCHEME — SET A

Tutor reveal · M = method, A = accuracy, B = independent mark, ECF = error carried forward

Q1

[8 marks]

B1 (a) 90 × 1000 = 90 000 m in 3600 s; 90 000 ÷ 3600 = 25 m/s

M1 (b) $E_k = \frac{1}{2} m v^2$

M1 $\frac{1}{2} \times 800 \times 25^2$ (= 400 × 625)

A1 250 000 J (2.5×10^5 J) — allow 250 kJ

B1 (c) E_k is proportional to v^2 (speed is squared)

B1 speed halves so E_k falls to one quarter (¼), not one half

B1 (d) thermal store of the brakes (and surroundings)

B1 by heating (friction)

Answer: (a) 25 m/s (b) 250 000 J

Q2

[7 marks]

M1 (a) $\frac{1}{2} \times 0.50 \times 3.0^2$

A1 2.25 J

B1 (b) speed doubles (2.0 → 4.0 m/s) but E_k increases from 1.0 to 4.0 J

B1 ×4, not ×2 — E_k is proportional to v^2, so not directly proportional to speed

M1 (c) $v^2 = 2E_k \div m = (2 \times 8.0) \div 0.50$

M1 $v^2 = 32$ then $v = \sqrt{32}$

A1 5.7 m/s (5.657… to 2 s.f.)

If the square root is omitted (v = 32): M1 M1 only.

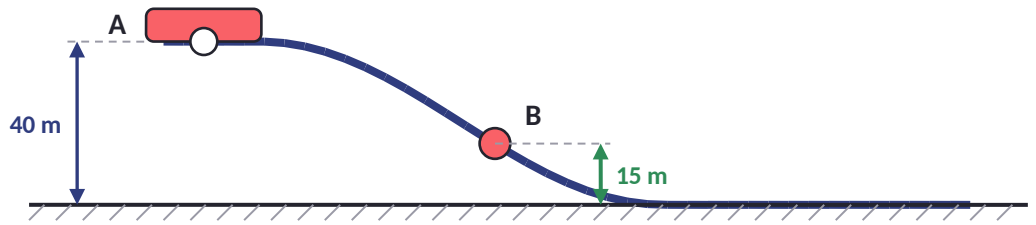
Answer: (a) 2.25 J (c) 5.7 m/s

EXAM PRACTICE — SET B

Gravitational potential energy · 14 marks in total · marks shown in [brackets] · mark scheme on the next slide

Q3 *Diagram · chained calculation* **[8 marks]**

A 500 kg roller-coaster car is released from rest at A. g = 10 N/kg.



(a) Calculate the gravitational potential energy of the car at A. [2]

(b) Calculate the energy transferred from the GPE store from A to B. [3]

(c) All this energy becomes kinetic energy. Calculate the speed at B. [3]

Q4 *Calculation · unit conversion* **[6 marks]**

A student of mass 60 kg climbs a staircase of 25 steps. Each step is 20 cm high. g = 10 N/kg.

(a) Show that the student climbs a height of 5.0 m. [1]

(b) Calculate the gravitational potential energy gained by the student. [2]

(c) The student now carries a 6.0 kg bag up the same stairs. Calculate the extra gravitational potential energy gained. [2]

(d) A box is lifted to twice the height. Its gravitational potential energy: A halves B stays the same C doubles D quadruples [1]

MARK SCHEME — SET B

Tutor reveal · M = method, A = accuracy, B = independent mark, ECF = error carried forward

Q3

[8 marks]

M1 (a) $E_p = m\ g\ h = 500 \times 10 \times 40$

A1 200 000 J (2.0×10^5 J)

M1 (b) uses the CHANGE in height: $\Delta h = 40 - 15 = 25$ m

M1 $\Delta E_p = 500 \times 10 \times 25$

A1 125 000 J (using 15 m gives 75 000 J: max 1 mark)

M1 (c) $\frac{1}{2} \times 500 \times v^2 = 125\,000$ (ECF from (b))

M1 $v^2 = 2 \times 125\,000 \div 500 = 500$

A1 v = 22 m/s (22.4… to 2 s.f.)

Answer: (a) 200 000 J (b) 125 000 J (c) 22 m/s

Q4

[6 marks]

B1 (a) 25 × 20 = 500 cm = 5.0 m (conversion must be shown)

M1 (b) $E_p = m\ g\ h = 60 \times 10 \times 5.0$

A1 3000 J (3.0×10^3 J)

M1 (c) $6.0 \times 10 \times 5.0$

A1 300 J (not 3300 J: the question asks for the EXTRA)

B1 (d) C (doubles) — E_p is directly proportional to h

Allow ECF in (b) from a wrong height in (a).

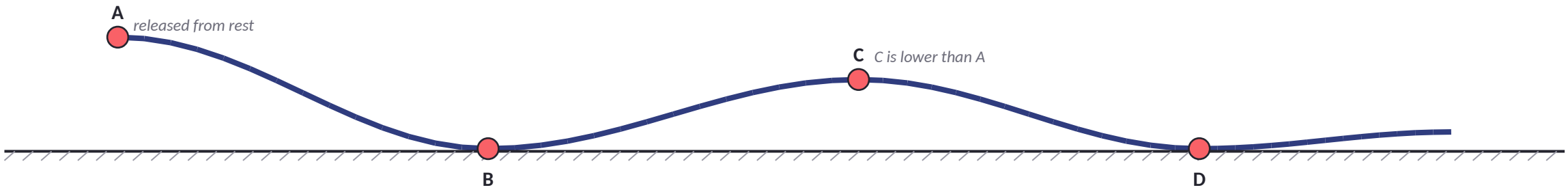
Answer: (b) 3000 J (c) 300 J (d) C

EXAM PRACTICE — 6-MARK QUESTION

Levels-based · 6 marks in total · marks shown in [brackets] · mark scheme on the next slide

Q5 *Extended response · levels* **[6 marks]**

*A roller-coaster car is pulled to the top of the first hill at A and released from rest. It runs down to B, up to the top of a lower hill at C, and then down to D. Describe the energy transfers as the car moves from A to D, and explain why the car cannot reach the height of A at C.



MARK SCHEME — 6-MARK QUESTION

Tutor reveal · M = method, A = accuracy, B = independent mark, ECF = error carried forward

Q5

[6 marks]

Level 3 (5–6) a clear, linked description of A→B, B→C and C→D using correct store names, AND a correct explanation of why C is lower than A (dissipation to the thermal store).

Level 2 (3–4) describes most transfers correctly, or explains the lower hill but with a missing / vague link (e.g. “friction” with no store named).

Level 1 (1–2) isolated correct statements (e.g. “GPE turns to KE”) with no sequence.

Indicative content: at A the car has maximum gravitational potential energy, no kinetic energy; A→B gravitational potential store empties and kinetic store fills (mechanically); at B kinetic store is maximum, GPE minimum; B→C kinetic store empties, GPE store fills; at C car is momentarily slow, GPE less than at A; C→D GPE empties, KE fills again. Friction and air resistance mean some energy is transferred by heating to the thermal store of the track, wheels and air (and some by sound) — dissipated — so the car has less energy than at A and cannot climb back to the same height.

Max 6 marks

Answer: **Levels-based: see descriptors**

Watch out — and what to remember

IN A NUTSHELL

- $E_k = ½ m v^2$ — speed is squared, don't drop the ½.
- $E_p = m g h$ — always check which g the question gives you.
- Rearrange directly for the squared equation; the triangle only helps for GPE.

COMMON SLIP-UPS

Squaring the wrong thing

In $E_k = ½ m v^2$, only v is squared — square it before multiplying by ½ and m, not the whole expression.

Forgetting to square-root

Rearranging for speed needs a final square-root step: $v = \sqrt{(2E_k \div m)}$, not just $2E_k \div m$.

Total height vs change in height

GPE only cares about the change in height — not the object's height above the ground at the start or the total drop of a multi-stage fall.

NEXT Deck 3 — Elastic potential energy & bringing it together →